LOGO

PathVLM-R1: A Reinforcement Learning-Driven Reasoning Model for Pathology Visual-Language Tasks

Jianyu Wu, Hao Yang, Yang Liu, *Member, IEEE*, Guibing He, Zhiyu Chen, Zihui Li, Yangyang Ma, Run Fang, Xinhua Zeng, Wei Zhou, *Senior Member, IEEE*

***Abstract*— The emergence of foundational models has catalyzed transformative advances in medical image analysis, yet their clinical deployment remains constrained by insufficient interpretability and opaque reasoning mechanisms. In pathology, where diagnostic decisions demand both accuracy and transparency, these limitations represent critical barriers to adoption. Current visual language models (VLMs) predominantly rely on supervised fine-tuning, which tends to capture superficial statistical correlations rather than develop genuine reasoning capabilities, which remains a fundamental shortcoming in high-stakes medical contexts. To address these challenges, we propose PathVLM-R1, a reasoning-enhanced visual language model for pathological image analysis. Building upon the Qwen2.5-VL-7B-Instruct foundation, PathVLM-R1 employs a meticulously designed two-stage post-training strategy: first injecting domain-specific pathological knowledge through supervised fine-tuning, then constructing robust reasoning capabilities via reinforcement learning with a novel dual-reward mechanism. The mechanism uniquely combines cross-modal process rewards with outcome accuracy rewards, ensuring both procedural rigor and diagnostic precision. Through group relative policy optimization, PathVLM-R1 achieves 65.50% accuracy on pathological visual question-answering tasks, surpassing both the baseline model by 13.2% and the substantially larger Qwen2.5-VL-32B model despite having fewer parameters. Furthermore, it demonstrates remarkable cross-modal transferability, achieving an average 17.3% improvement across out-of-domain medical imaging modalities including dermoscopy, chest CT, and retinal imaging, which underscore the effectiveness of the proposed PathVLM-R1 as a parameter-efficient, interpretable foundational architecture for diverse clinical applications.**

***Index Terms*— Pathology image analysis, visual-language model, reinforcement learning, visual question answering, foundational models.**

This work was supported by the AI-Assisted Pathological Diagnosis of Nephroblastoma Based on Virtual Staining and Multimodal Large Model, Medical-Engineering Cooperation Project of Fudan-Affiliated Hospital, "Dual Pairing" Program for Clinical & Basic Research of Children's Hospital of Fudan University, Dual Pairing Program of Children's Hospital of Fudan University, Construction of Digital & Intelligent Training Base for Pediatric Pathology, and Project of Shanghai Municipal Health Commission Shenkang. *(Jianyu Wu and Hao Yang contributed equally to this work.) (Corresponding authors: Yang Liu, Xinhua Zeng.)*

Jianyu Wu, Guibing He, Zhiyu Chen and Zihui Li are with the College of Intelligent Robotics and Advanced Manufacturing, Fudan University, Shanghai 200433, China (e-mails: {jywu24, gbhe23, zhiyuchen23, zihuili24}@m.fudan.edu.cn).

Hao Yang and Xinhua Zeng are with the College of Intelligent Robotics and Advanced Manufacturing, Fudan University, Shanghai 200433, China, and also with the Yiwu Research Institute of Fudan University, Yiwu, Zhejiang 322000, China (e-mails: haoyang24@m.fudan.edu.cn, zengxh@fudan.edu.cn).

Yang Liu is with the College of Electronic and Information Engineering, Tongji University, Shanghai 201804, China (e-mail: yang_liu@ieee.org).

Yangyang Ma is with the Department of Pathology, Children's Hospital of Fudan University, National Children's Medical Center, Shanghai 201102, China (e-mail: 12111240009@fudan.edu.cn).

Run Fang is with the Yiwu Research Institute of Fudan University, Yiwu, Zhejiang 322000, China (e-mail: fangrun@ywfudan.cn).

Wei Zhou is with the School of Computer Science and Informatics, Cardiff University, Cardiff, CF24 4AG, United Kingdom (e-mail: zhouw26@cardiff.ac.uk).

I. INTRODUCTION

The development of foundational models has ushered in a paradigm shift in medical image analysis, offering unprecedented capabilities for clinical decision support across diverse diagnostic modalities [1]–[3]. The large-scale architectures, pre-trained on vast multimodal datasets, have demonstrated remarkable proficiency in tasks ranging from lesion detection to automated report generation. However, their integration into clinical workflows aces a persistent challenge, i.e., the tension between model performance and interpretability, particularly in specialized domains such as pathology. Unlike general computer vision tasks, medical diagnosis requires not only accurate predictions but also transparent, verifiable reasoning processes that clinicians can scrutinize and trust.

Pathological diagnosis exemplifies this challenge with particular acuteness. The interpretation of histopathological images demands the recognition of subtle morphological variations, complex spatial relationships between cellular structures, and nuanced patterns in tissue architecture and staining characteristics. A diagnostic model that produces only classification labels, without articulating the visual evidence and logical steps underlying its conclusion, fails to meet the rigorous standards of pathological practice. Moreover, in high-stakes clinical scenarios where diagnostic errors can have profound consequences, the "black box" nature of many existing models creates justifiable hesitation among practitioners and regulatory bodies alike.

Current approaches to medical Visual Language Models (VLMs) predominantly employ Supervised Fine-Tuning (SFT)

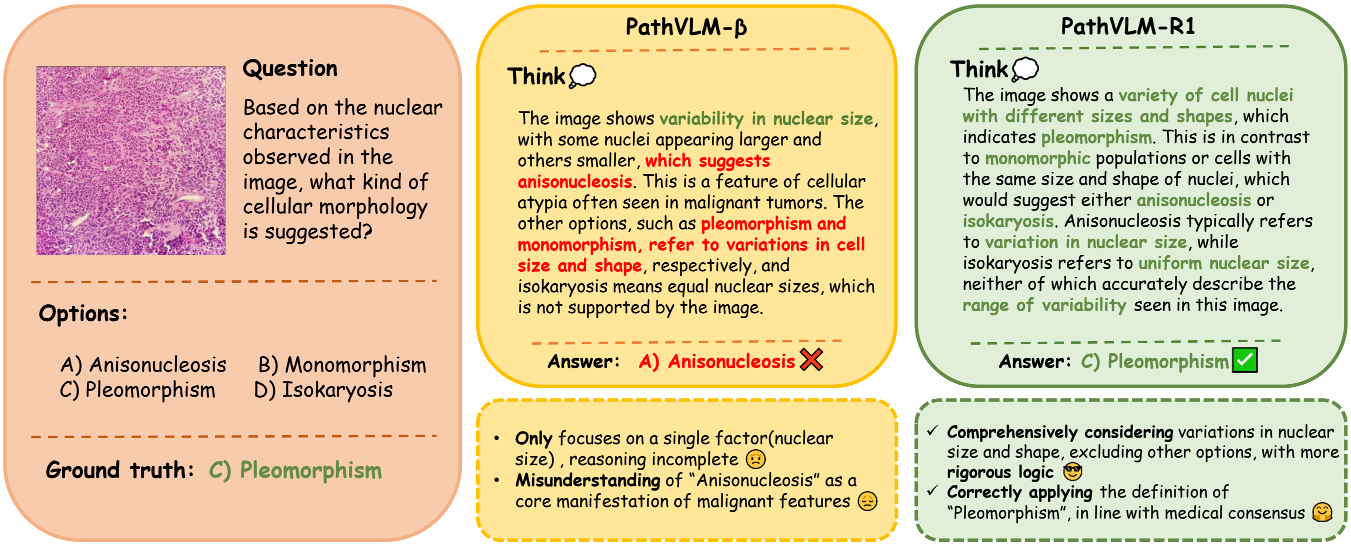

Fig. 1. Comparison of the final PathVLM-R1 model and its variant PathVLM-β. For the same pathology image question-answering task, the PathVLM-R1, which incorporates cross-modal procedural rewards, demonstrates superior rigor in reasoning and accuracy in medical knowledge compared to the PathVLM-β, which utilizes only accuracy rewards and format rewards. Specifically, in this example, PathVLM-β only focuses on differences in nuclear size while neglecting that "pleomorphism" includes multiple aspects of variation, such as nuclear shape and structure. In contrast, PathVLM integrates multiple nuclear features, accurately grasping the definition of "pleomorphism" with a more comprehensive thought process and stronger rigor.

paradigms [4], [5]. While SFT enables models to learn task-specific mappings from paired image-text data, it fundamentally optimizes for pattern matching rather than reasoning. Models trained exclusively through SFT tend to memorize correlations present in training data without developing the capacity to construct explicit, step-wise diagnostic rationales. This limitation is particularly problematic in medical contexts, where reasoning transparency is not merely desirable but essential for clinical acceptance. Furthermore, the heavy dependence on training data quality and coverage makes models vulnerable to performance degradation on rare conditions or atypical presentations—scenarios frequently encountered in real-world practice.

The limitations of supervised learning in fostering deep reasoning capabilities have motivated the exploration of alternative training paradigms. Reinforcement Learning (RL), widely employed in developing reasoning models for domains such as mathematics and logical inference, offers a compelling alternative by optimizing models to maximize cumulative rewards rather than merely mimicking training examples. By framing reasoning chain generation as a sequential decision-making process, RL encourages models to explore causal relationships between observations and conclusions, thereby constructing interpretable reasoning paths. Recent advances, such as DeepSeek-R1 [6], have demonstrated the potential of RL-based post-training to enhance reasoning in general domains. However, the direct application of these methodologies to specialized medical domains encounters several critical obstacles. First, foundational models lacking domain-specific medical knowledge yield inefficient and potentially erratic exploration during RL optimization. Without a solid grounding in pathological concepts, terminology, and diagnostic principles, the initial policy is essentially random, resulting in sparse reward signals that necessitate prohibitively large sample sizes for effective learning. Second, traditional RL approaches such as Proximal Policy Optimization (PPO) paired with Outcome Reward Models (ORM) [7], [8] focus exclusively on final outputs, overlooking the correctness and quality of intermediate reasoning steps. The outcome-centric paradigm is insufficient for medical applications, where process validity is as critical as result accuracy. Conversely, while PPO combined with Process Reward Models (PRM) can supervise reasoning steps, the approach incurs prohibitive costs for expert annotation of reasoning chains in specialized domains like pathology.

To overcome these challenges, we propose PathVLM-R1, a reasoning-enhanced VLM with a staged post-training framework. Our approach addresses the aforementioned limitations through two key innovations. First, we implement a two-phase training strategy: an initial knowledge injection phase where supervised fine-tuning on curated pathological datasets endows the model with foundational domain expertise, followed by a reasoning construction phase where reinforcement learning develops interpretable diagnostic reasoning capabilities. This sequential approach ensures that RL optimization operates on a knowledge-rich policy space, dramatically improving sample efficiency and training stability. Second, we introduce a novel dual-reward mechanism that holistically evaluates both reasoning process quality and outcome accuracy. Drawing inspiration from recent advances in Reinforcement Learning from AI Feedback (RLAIF) [9], we leverage the evaluative capabilities of advanced language models to generate scalable, high-quality process-level feedback without requiring expensive expert annotations. Our cross-modal process reward assesses reasoning chains along two critical dimensions: completeness of diagnostic logic (evaluating whether the model systematically analyzes visual features, considers differential diagnoses, and references appropriate medical knowledge) and correctness of medical reasoning (identifying violations of pathological principles, logical inconsistencies, or use of outdated diagnostic criteria). This process reward operates in tandem with outcome accuracy rewards that verify final diagnostic correctness. The integration of the complementary reward signals through Group Relative Policy Optimization (GRPO) [10] enables PathVLM-R1 to develop reasoning capabilities that are simultaneously interpretable and accurate.

The main contributions are summarized as follows:

- We present PathVLM-R1, the first VLM specifically optimized for pathological image analysis through a staged training paradigm that systematically builds from knowledge acquisition to reasoning capability construction. The model generates clinically interpretable reasoning explanations while achieving superior diagnostic accuracy.
- We transform the clinical imperative for transparent reasoning into learnable reward signals by quantifying reasoning quality through cross-modal process rewards and constraining diagnostic precision through outcome accuracy rewards. which operationalizes the joint optimization of evidence derivation and conclusion generation.
- PathVLM-R1 demonstrates that its training architectures can achieve performance surpassing larger models while requiring minimal reinforcement learning samples. With only 7 billion parameters and 1,000 RL training samples, it outperforms traditional SFT baselines and exhibits strong cross-modal transferability, highlighting its potential as a foundational model for clinical applications.

The remainder of this paper is organized as follows. Section II reviews related work on visual language models and reinforcement learning in post-training, positioning our contributions within the broader landscape of medical AI and foundational model research. Section III details our methodology, elaborating on the staged training framework, the GRPO algorithm, and the design of our dual-reward mechanism. Section IV presents comprehensive experimental evaluations on both in-domain pathological datasets and out-of-domain medical imaging modalities, demonstrating the effectiveness, efficiency, and generalization capabilities of PathVLM-R1. Finally, Section V concludes with a discussion of our findings, limitations, and directions for future research toward clinical validation and deployment.

II. Related Works

A. Visual Language Models

In recent years, general Visual Language Models such as LLama, Gemma, and QwenVL [11]–[16] have achieved a deep alignment of visual and linguistic representations through pre-training on large-scale image and text data, significantly advancing the field of multimodal understanding technology. The underlying Transformer [17] architecture and contrastive learning strategies provide effective means for capturing semantic associations between images and language [18]. These technological advancements offer important theoretical and practical foundations for constructing medical visual language models.

In the medical domain, researchers have drawn on the successful experience of general VLMs by transferring these models to medical images and clinical texts. Through supervised fine-tuning and instruction tuning, they adapt the models for specialized medical tasks. Traditional medical image-assisted diagnosis methods [19]–[21] require fine-tuning for each specific task. These specialized approaches range from classical computer vision techniques like template matching [22] for image segmentation [23] to more modern, yet still task-specific, Transformer architectures for classifying whole-slide images. In contrast, the new learning paradigm using VLMs can effectively leverage web data and zero-shot prediction, allowing VLMs to identify new objects with few or no image samples [24]. For example, the LLaVA-Med [4] models leverage large volumes of medical image-text pairs for further training, achieving significant progress in tasks like automated image report generation and visual question-answering. Furthermore, the VLM paradigm is already expanding beyond 2D modalities, with models like Med3DVLM [5] demonstrating its efficacy for analyzing 3D medical images. Despite these advancements, current medical visual language models still face several challenges:

Firstly, many medical visual language models primarily employ methods such as supervised fine-tuning, unsupervised pre-training, and contrastive learning. These methods aim to capture the statistical correlations and surface patterns between images and text. The commonly used training data typically involve paired sets of images and final diagnostic reports, clinical records, or radiology reports. Such data often only record diagnostic outcomes or report summaries, lacking detailed descriptions of the intermediate reasoning steps involved in the diagnostic process. This limitation affects existing medical visual language models in terms of inference capability and clinical interpretability, making it difficult to construct clear and transparent medical reasoning chains that could provide deep diagnostic insights.Recently, researchers have attempted to enhance model reasoning performance on medical tasks using reinforcement learning with GRPO as the optimization strategy. For instance, the MedVLM-R1 [25] model has surpassed baseline methods that use supervised fine-tuning in tests involving MRI, CT, and OCT modalities. However, it failed in tests on pathology modalities. Moreover, its approach of directly applying reinforcement learning to base models lacking fundamental knowledge has resulted in relatively limited performance improvements.

Secondly, since the training data is often restricted to specific domains and imaging modalities, the models have limited generalization capability when confronted with diverse clinical scenarios.

B. Reinforcement Learning in the Post-training Phase

Reinforcement learning (RL) has emerged as a powerful methodology for optimizing sequential decision-making processes across various domains, including healthcare. Beyond image analysis, RL has been successfully applied to complex clinical policy management, such as optimizing the switching strategy for noninvasive ventilation in patients [26]. This demonstrates the capability of RL to learn effective policies from interactions in environments where optimal decisions

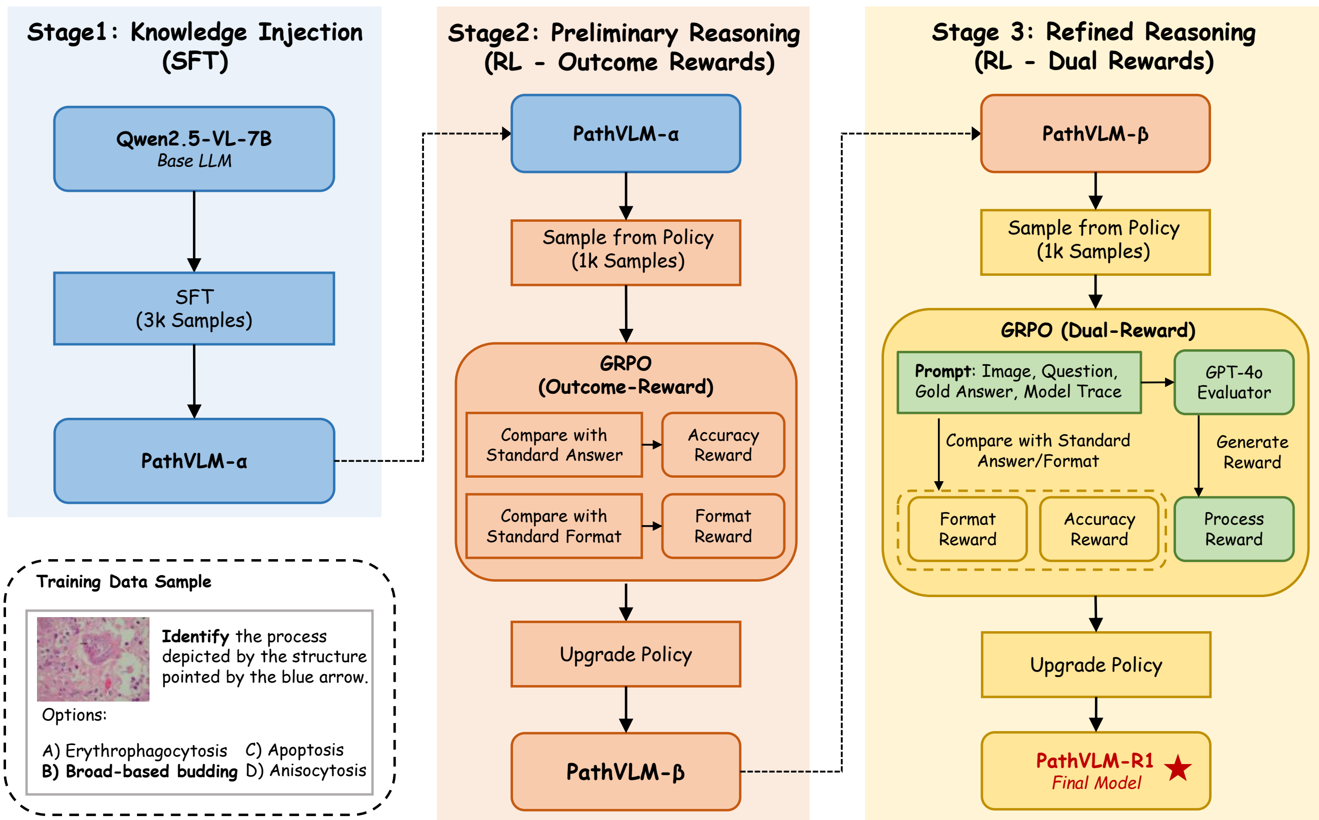

Fig. 2. The PathVLM-R1 staged training pipeline is depicted in the accompanying diagram, illustrating the progressive development of the model from foundational knowledge acquisition to refined reasoning capabilities. In Stage 1 (Knowledge Injection), the Qwen2.5-VL-7B base large language model (LLM) undergoes supervised fine-tuning (SFT) on a 3,000-sample pathology dataset to acquire essential pathological knowledge, resulting in the intermediate model, PathVLM-α. In Stage 2 (Preliminary Reasoning), PathVLM-α is further optimized using Group Relative Policy Optimization (GRPO), where the reward signals are limited to outcome-based metrics—namely, Accuracy Reward and Format Reward. This stage is designed to establish initial reasoning capabilities, producing PathVLM-β. Finally, Stage 3 (Refined Reasoning) involves an additional GRPO-based refinement of PathVLM-β, introducing a Dual-Reward mechanism that integrates both outcome-based rewards and a Process Reward. The Process Reward is dynamically generated by a GPT-4o Evaluator, which assesses a comprehensive prompt containing the input image, the posed question, the reference answer (Ref-A), and the model-generated reasoning trace (Model Trace). The evaluator's assessment of the reasoning quality guides the policy updates, enabling the model to produce more robust and coherent reasoning chains. This multi-stage procedure culminates in the final PathVLM-R1 model, which combines foundational knowledge with advanced reasoning capabilities.

are not immediately obvious. Framing the generation of a coherent reasoning chain as a sequential decision-making problem—where the model must select the best subsequent token at each step to maximize a final reward—aligns perfectly with the principles of RL. This perspective motivates our application of RL in the post-training phase of large models. Reinforcement Learning from Human Feedback (RLHF) [27] is a common practice during the post-training phase of large models [28]–[30]. RLHF is a methodological approach that integrates human feedback into the reinforcement learning process. By collecting and analyzing human preferences or ratings for different behaviors, RLHF trains a reward model that simulates human preferences. Subsequently, using reinforcement learning algorithms such as Proximal Policy Optimization (PPO), this approach employs the initial model as a policy and uses the reward model's ratings as optimization signals to iteratively refine the model outputs, aligning them more closely with human values. A seminal work is InstructGPT [31], which employs Reinforcement Learning from Human Feedback (RLHF) to enhance the model's ability to adhere to instructions given in text prompts, thereby reducing harmful outputs and improving factual accuracy.

There are primarily two types of reward models: the Outcome Reward Model (ORM) and the Process Reward Model (PRM). The ORM focuses on the final output, rating and evaluating the overall result generated by the model, while the PRM scores each step in the generation process. Training these reward models typically relies on a substantial amount of human-labeled data. However, in fields such as medicine, where high-quality annotated data is scarce, the difficulty of training reward models increases significantly.

III. METHODS

In the post-training phase for pathology tasks, we employ carefully crafted training strategies to enhance the model's reasoning capability and performance in these tasks. Below, we will introduce the specific optimization algorithms and training processes.

A. Group Relative Policy Optimization

In the medical field, the sensitivity of data, barriers due to specialized knowledge, and the high cost of annotations make acquiring high-quality reasoning process labels extremely difficult. This presents significant challenges to traditional reinforcement learning methods that rely on explicit reward models or value functions, such as Proximal Policy Optimization (PPO). In this context, GRPO emerges as a highly promising solution.

Group Relative Policy Optimization is an efficient reinforcement learning algorithm designed to enhance model reasoning by optimizing policy gradients through group-based relative advantage estimation and regularization, addressing limitations of traditional methods like PPO. The algorithm operates on a problem distribution $P(Q)$, where each problem q is processed by an old policy $\pi_{\theta^{\text{old}}}$ to generate G candidate responses $\{o_i\}_{i=1}^{G}$, with π_θ denoting the current policy to be optimized and π_{ref} serving as a frozen reference policy to constrain updates via KL divergence. For each candidate response o_i, a rule-based reward r_i is computed, and relative advantage is derived through group-wise normalization:

$$A_i = \frac{r_i - \operatorname{mean}(r_1, \ldots, r_G)}{\operatorname{std}(r_1, \ldots, r_G)} \tag{1}$$

which measures the quality of o_i relative to the group. The objective function of GRPO balances policy improvement and stability as:

$$\begin{aligned} \mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E}&\left[q \sim P(Q),\ \{o_i\}_{i=1}^{G} \sim \pi_{\theta_{\text{old}}}(O \mid q)\right] \\ &\frac{1}{G}\sum_{i=1}^{G}\Bigg\{\min\Bigg[\frac{\pi_\theta(o_i \mid q)}{\pi_{\theta_{\text{old}}}(o_i \mid q)}A_i, \\ &\operatorname{clip}\left(\frac{\pi_\theta(o_i \mid q)}{\pi_{\theta_{\text{old}}}(o_i \mid q)},\ 1-\varepsilon,\ 1+\varepsilon\right)A_i\Bigg] \\ &-\beta\ \mathbb{D}_{\text{KL}}\left[\pi_\theta \| \pi_{\text{ref}}\right]\Bigg\} \end{aligned} \tag{2}$$

where G is the number of candidate responses per problem, $\frac{\pi_\theta(o_i|q)}{\pi_{\theta_{\text{old}}}(o_i|q)}$ is the policy ratio reflecting the change in response generation probability, $\operatorname{clip}(\cdot)$ with hyperparameter ϵ stabilizes training by limiting extreme updates, and $\beta\mathbb{D}_{\text{KL}}(\pi_\theta \| \pi_{\text{ref}})$ ensures π_θ does not deviate excessively from the reference policy π_{ref}, with β controlling the regularization strength. Unlike PPO, which relies on a value function for advantage estimation, GRPO leverages group statistics to compute A_i, eliminating the need for a separate critic model and reducing computational costs by 50%. The algorithm iteratively updates π_θ by maximizing this objective, synchronizing parameters with $\pi_{\theta^{\text{old}}}$ for the next iteration, thus enabling efficient exploration of solution spaces to enhance reasoning capabilities, as demonstrated by notable improvements in benchmarks like MATH.

B. Redesigning Reward Mechanisms for Interpretability

During the training of Deepseek-R1 [6], distinct reward mechanisms were tailored for different types of tasks. For tasks with well-defined standard answers or those that can be mechanically verified (such as mathematics, code, and logical reasoning), rule-based reward functions were utilized for optimization. For tasks lacking explicit rules or requiring subjective evaluation (such as creative writing and open-ended questions), a reward model capturing human preferences was introduced to address the limitations of rule-based rewards in unstructured tasks. This aims to balance reasoning capability with adaptability in general scenarios.

Similarly, to fulfill the requirements of process interpretability and result accuracy in reasoning within medical contexts, we designed a reward mechanism composed of cross-modal process rewards and result accuracy rewards. This mechanism ensures logical supervision of the reasoning process and constraints on result accuracy. The detailed workflow of this dual-reward mechanism is illustrated in Fig. II-A, where the model's output is bifurcated to generate a Process Reward that evaluates the reasoning quality and Outcome Rewards that verify the final answer's correctness.

The result accuracy rewards follow the design of Deepseek-Math [10], comprising format rewards and accuracy rewards. The format reward requires the model to produce outputs in a specified format: explicitly articulating the reasoning process within a ¡think¿ tag and providing the final answer within an ¡answer¿ tag. Outputs that meet the format specifications receive a score of 1. In accuracy evaluation, the model must present a clear letter option and content within the ¡answer¿ tag, receiving a score of 1 only for completely correct answers.

However, test results based solely on format and accuracy rewards indicate that the quality of the generated reasoning chains is often suboptimal, frequently exhibiting logical leaps or misuse of medical arguments. Despite this, even low-quality reasoning chains can sometimes lead to correct results, which is detrimental to the model's ability to find effective strategies. We realized that for open-ended tasks such as medical reasoning, accuracy rewards alone are insufficient. External models are needed to supervise the reasoning process. Given the scarcity of high-quality reasoning process data in the medical field, we referenced Reinforcement Learning with AI Feedback (RLAIF) [9], [32], [33] and DeepSeek-R1 [6]'s training strategies to design a segmented approach.

We input images, questions, standard answers, model outputs, and prompts into the GPT-4o model, which evaluates them to generate cross-modal rewards. In the prompts, we specified the GPT-4o's evaluation criteria and response format. To prevent GPT-4o from returning biased formatting or irrelevant information, we designed a detailed process for handling its feedback.

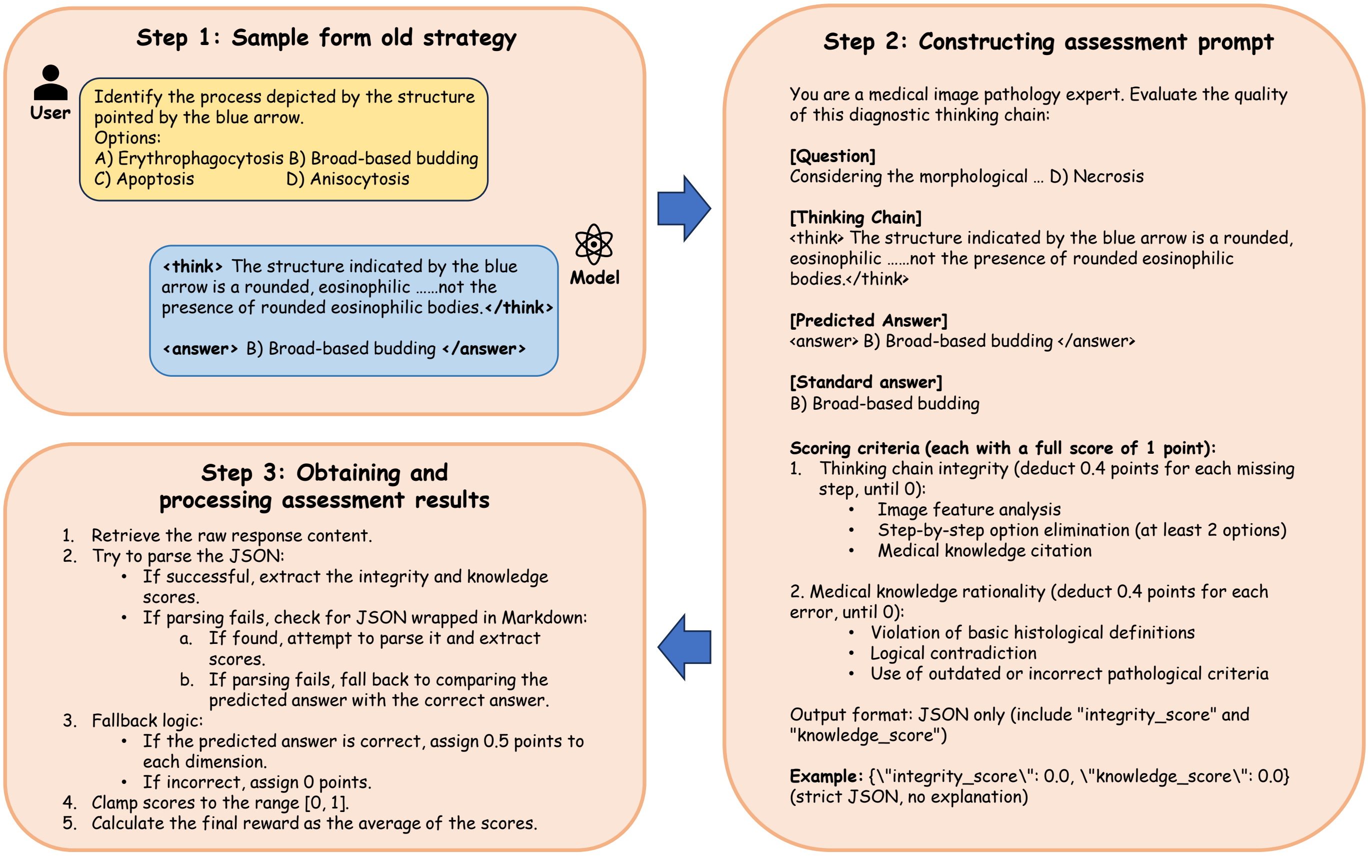

Fig. 3. Pipeline for cross-modal procedural loss. First, sampling is performed on the old policy to obtain the model's generated thought process and final answers. These, along with the questions, generated content, and evaluation criteria, are then given to GPT-4o for review in terms of reasoning completeness and knowledge correctness, obtaining an Integrity score and Knowledge score. Lastly, GPT-4o's feedback is processed, including error handling and score normalization, using the average of the Integrity score and Knowledge score as the final reward.

Evaluation criteria include completeness of the reasoning chain and reasonableness of medical knowledge, each with a full score of 1 point. Completeness of the reasoning chain encompasses the analysis of image features, stepwise elimination of options, and reference to medical knowledge, with a deduction of 0.4 points for each missing step, with a minimum score of 0. Medical knowledge reasonableness involves breaches of basic histological definitions, logical contradictions, and the use of outdated or inappropriate pathological standards, with 0.4 points deducted for each infraction, also with a minimum score of 0. Fig.III-A illustrates the detailed evaluation process.

To validate the stability and reliability of our GPT-4o based reward scorer, we conducted a consistency analysis by having it score the same 500 reasoning chains in three independent runs. As visualized in Fig.4, the analysis reveals nearly identical score distributions and strong pairwise correlations (Spearman's $\rho > 0.78$) across all runs. This result confirms that our evaluator provides a consistent and robust reward signal, which is a crucial prerequisite for effective reinforcement learning.

IV. Experiment and Results

A. Experimental Setup

Dataset This study utilized the PathMMU [34] dataset for experiments, which comprises 33,428 multimodal multiple-choice questions (Q&A) and 24,067 pathological images. We selected the PubMed and EduContent subsets, totaling 5,385 entries for training and testing. Among these, 3,000 were used for training the Supervised Fine-Tuning base model, 1,000 for reinforcement learning training, and the remaining 1,385 for testing. Furthermore, to verify the model's generalization capability, we selected ChestCT [35], ISIC2020 [36], Retinal OCT-C8 [37], and Diabetic Retinopathy [38] from OmniMedVQA [39] as out-of-domain data for testing. The detailed breakdown of our data utilization for each experimental phase is provided in Table I. The out-of-domain datasets from OmniMedVQA were chosen to represent a diverse array of medical imaging modalities, including computed tomography (ChestCT), dermoscopy (ISIC2020), and retinal imaging (OCT and fundus photography). This diversity provides a robust testbed for assessing the model's ability to transfer its learned reasoning skills from the microscopic world of pathology to visually and contextually distinct clinical domains.

Evaluation Metrics To thoroughly assess the performance of our models, we employed a multi-faceted evaluation strategy. Beyond traditional accuracy for the final answer, we utilized a comprehensive set of human-aligned metrics to evaluate the quality of the generated reasoning chains. These metrics, scored on a 0-1 scale by a GPT-4o-based evaluator, encompass:

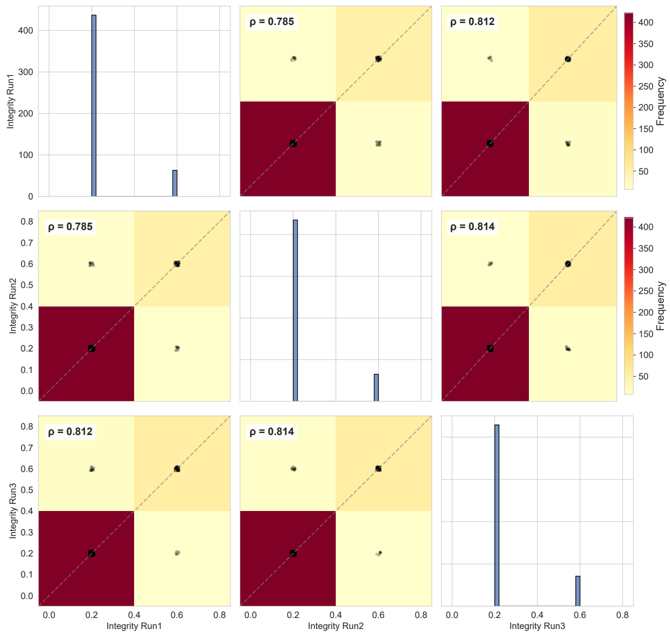

Fig. 4. Consistency Analysis of GPT-4o Integrity Scores across Three Independent Runs. This correlation matrix illustrates the high scoring consistency of the GPT-4o evaluator for the reasoning integrity dimension.Diagonal Panels: The histograms show the score frequency distribution for each of the three independent runs. Their nearly identical shapes demonstrate a highly consistent scoring pattern applied in each session.Off-Diagonal Panels: The 2D density heatmaps visualize the pairwise correlations between the runs. The concentration of data along the diagonal (dashed line) visually confirms the strong agreement.Quantitative Analysis: The Spearman's rank correlation coefficient (ρ) is annotated for each pair, with all values exceeding 0.78, indicating a strong and reliable positive correlation between the independent evaluations.

TABLE I
OVERVIEW OF THE PATHMMU DATASET SUBSETS USED IN THIS STUDY.

Statistical Item	**Quantity/Description**
Total Images in PathMMU	24,067
Total Q&A Pairs in PathMMU	33,428
Q&A Pairs Used in This Study	**5,385 (from PubMed & EduContent)**
For SFT Training	3,000
For RL Training	1,000
For Performance Testing	1,385

- **R-ACC:** Evaluates if the reasoning logically supports the final answer, is evidence-based, and sufficiently conclusive. Concise and effective reasoning is prioritized.
- **K-ACC:** Assesses the accuracy of medical/pathological terminology, adherence to basic medical principles, and appropriate use of diagnostic criteria.
- **Rigor:** Examines the natural flow of reasoning steps, absence of logical leaps or unsupported jumps, and the necessity of conclusions from premises.
- **Professionalism:** Evaluates the use of appropriate medical terminology and language suitable for experts, avoiding casual expressions.
- **Clarity:** Measures the ease of understanding, organization, and adequate explanation of concepts within the reasoning.
- **Conciseness:** Focuses on the conciseness and directness of the reasoning, penalizing unnecessary verbosity.

The primary objective of these detailed metrics is to quantify the nuanced aspects of diagnostic reasoning, beyond just the final outcome.

Implementation Details Experiments were conducted on a setup with four H20 GPUs (96GB of memory). The training base is Qwen2.5-VL-7B-Instruct [13], with the visual encoder frozen. The parameter G is set to 4, and both the maximum prompt length and the maximum generation length are set to 1024.

The experimental procedure consists of the following three phases:

- Step 1: SFT Phase to Obtain the Alpha Model Utilizing 3,000 data points to train the Qwen2.5-VL-7B-Instruct model for 18,000 steps until performance stabilizes, resulting in the new base model, Alpha.
- Step 2: Reinforcement Learning Phase to Obtain the Beta Model This is the first step in constructing reasoning capabilities. Using 1,000 newly introduced data points and applying rewards for accuracy and formatting only, Alpha is optimized for 2,000 steps until performance stabilizes, resulting in the Beta model.
- Step 3: Final Optimization Using Cross-Modal Process Rewards Continuing with the same 1,000 data points, combined with cross-modal process rewards and outcome accuracy rewards, the Beta model is further optimized for 2,000 steps until performance stabilizes, ultimately yielding the PathVLM-R1 model. We aim for the model to be faithful to the data distribution of the training set while avoiding potential reward hacking issues, thus choosing to further refine the reasoning process after initial model exploration.

Fig.5 presents the model variants obtained during the training process.

To ensure the rigor of the experiment, we established the following control groups:

- Directly using 1,000 data points on the Qwen2.5-VL-7B-Instruct with accuracy and format rewards, optimized via GRPO, termed as model Epsilon, along with a supervised fine-tuning control group called model Delta.
- Continued SFT training on the Alpha model, termed as model Gamma.

B. Experimental Results

In this section, we present a systematic evaluation of PathVLM-R1 to validate the efficacy of our proposed training framework. We conducted a series of experiments on both in-domain and out-of-domain datasets, assessing the model's diagnostic accuracy, reasoning capabilities, generalization performance, and interpretability.

In-domain Performance and Staged Training Analysis Our primary experiments were designed to quantify the contribution of each stage in our "knowledge injection followed by reasoning construction" pipeline. The results, summarized in Table II and visualized in Fig.6, demonstrate a clear and progressive enhancement of the model's performance.

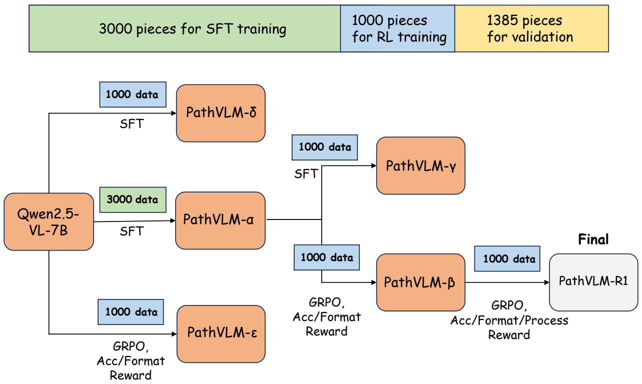

Fig. 5. Various model variants generated during stage-wise training. Through the comparison of various model variants generated at each stage, this pipeline empirically validates the effectiveness of our staged training methodology. The entire training set is divided into disjoint segments in the form of 3000, 1000, 1385, used for supervised fine-tuning, reinforcement learning, and its control group, as well as the final performance testing process, respectively. Using Qwen2.5-VL-7B as the base, the model Alpha is obtained with 3000 data points for supervised fine-tuning. Delta and Epsilon are obtained using 1000 data points for supervised fine-tuning/reinforcement learning. Gamma and Beta are obtained by continuing supervised fine-tuning/reinforcement learning with an additional 1000 data points on Alpha. Finally, adding cross-modal procedural losses to Beta results in the final model PathVLM-R1.

The baseline model, Qwen2.5-VL-7B, achieved an initial accuracy of 51.55% on the in-domain test set. In the first stage, we performed supervised fine-tuning (SFT) on 3,000 pathological data pairs to inject foundational domain knowledge. This resulted in the PathVLM-Alpha model, which saw its accuracy significantly increase to 57.59%. This outcome underscores the criticality of SFT as a preliminary step to equip the model with the necessary pathological concepts and terminology.

Building upon the knowledge-infused PathVLM-Alpha, we explored two distinct paths for the second stage using 1,000 new data samples. Continuing with SFT yielded the PathVLM-Gamma model with an accuracy of 59.92%. In contrast, applying our reinforcement learning strategy with only outcome-based rewards (accuracy and format) produced the PathVLM-Beta model, which achieved a substantially higher accuracy of 63.89%. As illustrated by the steeper performance curve in Figure 6, this demonstrates that once a knowledge base is established, reinforcement learning is a more potent strategy than SFT for cultivating sophisticated reasoning abilities. The final and most crucial optimization step involved incorporating our novel cross-modal process rewards. By fine-tuning PathVLM-Beta with this dual-reward mechanism, we developed the final PathVLM-R1 model, which attained a peak accuracy of 65.50%. The detailed accuracy improvements for each model variant at every stage of our pipeline are comprehensively presented in Table V.

To further validate our staged approach, we established control groups. The PathVLM-Epsilon model, trained by applying RL directly to the 7B base model, achieved an accuracy of only 56.10%. Although this outperformed its direct SFT counterpart (PathVLM-Delta), the model's performance was unstable, likely due to a lack of domain knowledge that led it to learn superficial strategies. These findings highlight the importance of initial knowledge injection. The superiority of our final method is evident as PathVLM-R1 surpassed the supervised baseline (PathVLM-Gamma) by 7.96%, whereas directly applying RL from the start resulted in a much smaller increase.

Our findings contrast with other reasoning models for medical imaging, such as MedVLM-R1, which employ a rule-based reward function directly on a base model optimized via GRPO. This direct approach has yielded suboptimal results, and MedVLM-R1 notably failed in tests on pathology images. Our experimental results strongly suggest that our post-training strategy of ”pathological knowledge infusion followed by reasoning capability construction” is a more effective and efficient paradigm for achieving superior performance in complex medical domains.

In addition to accuracy, we analyzed the average output token count across different models, as shown in Table III. We observed that the initial 18,000 steps of SFT led to a notable reduction in the output token count of PathVLM-Alpha. However, this issue of brevity was effectively alleviated by the subsequent 2,000-step optimization with cross-modal process rewards, which encouraged the generation of more detailed and complete reasoning chains in the final PathVLM-R1 model.

Beyond quantitative metrics, it is crucial to verify that the model's reasoning is grounded in relevant visual evidence, a key factor for clinical trust. To this end, we visualized the model's cross-modal attention to understand where it ”looks” while generating its response. Fig.7 presents a compelling case study where the attention heatmap confirms that PathVLM-R1 correctly focuses on the specific pathological features mentioned in the prompt—namely, the peripheral insert with immunohistochemical staining. This strong alignment between the model's visual focus and its textual reasoning provides clear evidence of its interpretability and robust diagnostic capability.

Analysis of Out-of-domain Data Results To assess the model's ability to transfer learned knowledge, we evaluated PathVLM-R1 on four out-of-domain medical imaging modalities not present in its training set. The results are detailed in Table IV.

PathVLM-R1 demonstrated remarkable generalization capability on the ISIC 2020 dermoscopy dataset, achieving an accuracy of 80.00%. This represents a 23.41% improvement over the Qwen2.5-VL-7B baseline model. This result is particularly significant because our training data consisted of microscopic skin pathology slides, whereas the ISIC 2020 test data contains macroscopic images of surface lesions. The successful knowledge transfer between these different scales indicates that our model has learned the underlying pathological principles rather than just superficial image features, showcasing its potential as a robust and adaptable pathology model. While its performance on other modalities like Chest CT and Retinal OCT did not exceed models specifically trained on such data, it consistently outperformed the SFT baseline, further evidencing its impressive generalization potential.

Interpretability and Visual Grounding For any model

TABLE II
IN-DOMAIN COMPARISON OF MODELS WITH DIFFERENT PARAMETER SCALES (ACC. REPRESENTS ACCURACY).

Model	Samples	ACC	R-ACC	K-ACC	Rigor	Professionalism	Clarity	Conciseness
Large-parameter models								
Doubao-1.5-vison	500	45.80	0.5253	0.5719	0.5399	0.8715	0.8395	0.7715
Qwen-VL-Plus	500	46.60	0.5326	0.5791	0.5417	0.8729	0.8381	0.7196
DeepSeek-VL2	500	49.40	0.5217	0.5777	0.5261	0.8644	0.8291	0.7528
Claude-3.5-Haiku	500	54.20	0.6022	0.6488	0.6118	0.9000	0.8582	0.7886
Grok-4-Fast	500	55.80	0.6219	0.6718	0.6298	0.9048	0.8549	0.7837
LLaMA3.2-90B	500	63.80	0.6352	0.7100	0.6521	0.9144	0.8522	0.7176
Claude-4.5	500	**64.80**	**0.7000**	**0.7448**	**0.7108**	**0.9285**	**0.8857**	**0.8102**
Comparable-parameter models								
Model	Samples	ACC	R-ACC	K-ACC	Rigor	Professionalism	Clarity	Conciseness
LLaMA3.2-11B	500	23.40	0.3681	0.3983	0.3770	0.8330	0.8090	0.6889
MedVLM-R1	500	42.60	0.4263	0.5090	0.4410	0.7840	0.7361	0.6780
MedGemma-4B	500	44.40	0.4862	0.5524	0.4935	0.8463	0.7980	0.7154
Qwen2.5-VL-3B	500	44.40	0.5016	0.5508	0.5067	0.8616	0.8267	0.7567
Qwen2.5-VL-7B	500	51.80	0.5799	0.6276	0.5855	0.8879	0.8479	0.7728
Lingshu-7B	500	55.60	0.6248	0.6612	0.6284	**0.9056**	**0.8684**	0.7992
InternVL3-8B	500	57.20	0.6152	0.6569	0.6184	0.8986	0.8683	0.8130
Ours(PathVLM-R1)	500	**65.50**	**0.6323**	**0.6950**	**0.6373**	0.9040	0.8663	**0.8228**

TABLE III
AVERAGE NUMBER OF TOKENS OUTPUTTED BY DIFFERENT MODELS ON IN-DOMAIN DATA.

Model	Average number of tokens
Qwen2.5-VL-7B	210.04
PathVLM-β-7B	80.31
PathVLM-ϵ-7B	220.07
PathVLM-R1	94.99

TABLE IV
OUT-OF-DOMAIN COMPARISON OF DIFFERENT MODELS AND THEIR ACCURACIES.

Model	Dataset	Accuracy (%)
Qwen2.5-VL-3B	Chest CT	38.60
	ISIC 2020	34.59
	Retinal OCT-C8	**60.40**
	Diabetic Retinopathy	64.20
Qwen2.5-VL-7B	Chest CT	48.60
	ISIC 2020	56.59
	Retinal OCT-C8	57.19
	Diabetic Retinopathy	**70.80**
HuatuoGPT-Vision-7B	Chest CT	**51.00**
	ISIC 2020	34.20
	Retinal OCT-C8	48.19
	Diabetic Retinopathy	62.00
PathVLM-γ	Chest CT	40.60
	ISIC 2020	45.40
	Retinal OCT-C8	45.80
	Diabetic Retinopathy	32.00
Ours (PathVLM-R1)	Chest CT	44.00
	ISIC 2020	**80.00**
	Retinal OCT-C8	49.20
	Diabetic Retinopathy	59.80

intended for clinical application, ensuring its decisions are transparent and trustworthy is paramount. To this end, we qualitatively analyzed PathVLM-R1's reasoning process by visualizing its cross-modal attention mechanisms. The heatmap generation process is technically intricate. First, we programmatically construct a mask to isolate the indices of pure text tokens; this is achieved by identifying the index ranges corresponding to image tokens (demarcated by $<|\text{vision\_start}|>$ and $<|\text{vision\_end}|>$) and excluding them, while also filtering out special control symbols. With this text-position mask, we proceed to the attention matrices within each transformer layer. For each layer, we average the attention matrix across all heads, then select the attention weights corresponding to the flow from the text positions to all tokens in the sequence. By averaging these weights along the text dimension, we derive a single vector representing the collective attention from the entire text block to every token (both text and image). We then isolate the segment of this vector that corresponds to the image patches, which yields a final vector quantifying the overall textual focus on each part of the image. This vector is subsequently normalized, reshaped into a two-dimensional grid, and finally interpolated and overlaid onto the original input image.

As shown in Fig.7, the model was tasked with identifying a staining pattern within a specific "peripheral insert" of the pathology image. The attention heatmap reveals that the model's high-intensity focus (yellow/red regions) is precisely localized over this specific area of interest. This strong visual grounding confirms that the model's textual reasoning—identifying a "membranous distribution"—is directly based on the correct visual evidence from the image. This alignment between the model's visual focus and its generated explanation provides clear evidence of its interpretability and enhances the trustworthiness of its diagnostic process.

V. DISCUSSION

Interpretation of Principal Findings Our primary finding is that a staged training strategy is crucial for adapting large models to specialized domains like pathology. We observed that directly applying reinforcement learning to a base model

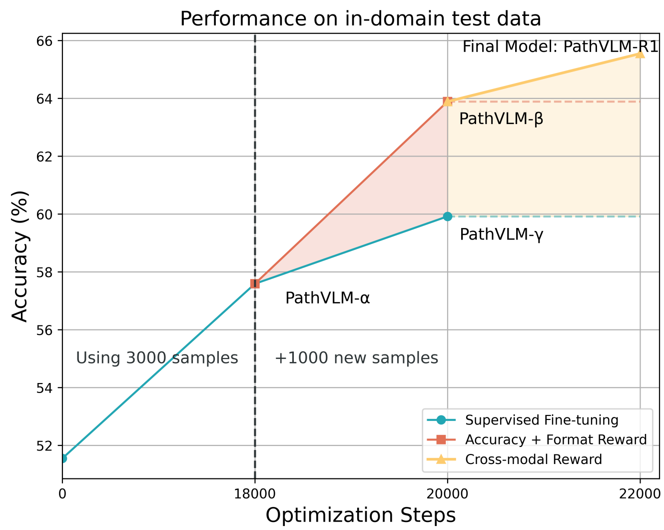

Fig. 6. Changes in model accuracy across different optimization stages. The model's performance is significantly improved across all three training steps.

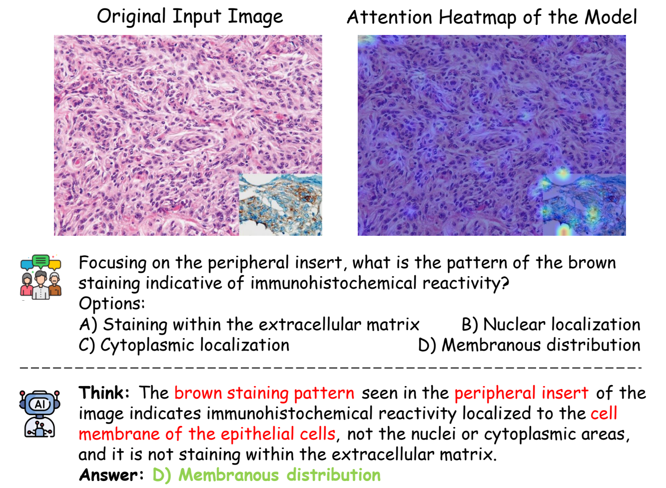

Fig. 7. Visual Grounding of PathVLM-R1's Reasoning via Attention Heatmap. This figure provides a qualitative example of the model's interpretability.(Left Panel) The input provided to the model, which includes the original pathology image, a question focused on the peripheral insert, and the ground truth answer.(Right Panel) The model's output, including the generated reasoning chain (think) and the correct final answer (answer). The attention heatmap, generated by aggregating attention scores from the model's transformer layers, is overlaid on the original image. The heatmap's high-intensity regions (yellow/red) are precisely localized over the peripheral insert, confirming that the model bases its reasoning—specifically identifying the "membranous distribution"—on the correct visual evidence. This demonstrates a strong cross-modal alignment and enhances the trustworthiness of the model's diagnostic process.

(PathVLM-Epsilon) yielded unstable and suboptimal performance, even though it surpassed its direct SFT counterpart (PathVLM-Delta). This supports our hypothesis that without a foundational layer of domain knowledge, the model's exploration during RL can be inefficient and may lead to the learning of superficial strategies. The initial SFT phase, which produced PathVLM-Alpha, proved to be an indispensable step for knowledge injection, establishing a robust foundation upon which complex reasoning capabilities could be built. Furthermore, the significant performance increase from PathVLM-Beta to our final PathVLM-R1 model highlights the superiority of supervising the reasoning process itself, not merely the final outcome. By incorporating a cross-modal process reward, we guided the model to develop reasoning chains that are more logically sound and aligned with medical knowledge, directly addressing the "black box" challenge in clinical AI. The exceptional parameter efficiency of our approach, where our 7B model outperformed a 32B model, suggests that sophisticated training architecture is a more effective path to specialized expertise than sheer model scale.

Interpretation of Clinical Implications The clinical implications of PathVLM-R1 extend significantly beyond its accuracy improvements, primarily by addressing the critical barrier of clinical trust in medical AI. Unlike opaque "black box" models, our approach yields an auditable "glass box" partner. The model generates explicit, interpretable reasoning chains , allowing clinicians to scrutinize the diagnostic logic (as shown in Fig. I) . This trust is further solidified by the model's strong visual grounding; attention heatmaps (Fig. 7) confirm that its textual reasoning—such as identifying a "membranous distribution"—is directly anchored to the correct visual evidence, namely the "peripheral insert" in the image. This verifiable reasoning process is directly coupled with enhanced diagnostic utility. As demonstrated in the differentiation of "Pleomorphism" from "Anisonucleosis" (Fig. I), PathVLM-R1 succeeds by comprehensively evaluating multiple visual features (both nuclear size and shape), whereas the baseline model failed by focusing on only a single factor. The model's resulting capability suggests its potential as a powerful decision-support tool to reduce diagnostic variability and improve accuracy, particularly in morphologically ambiguous cases that demand nuanced interpretation. Furthermore, the model's remarkable generalization, evidenced by its 80.00% accuracy on the ISIC 2020 dermoscopy dataset, carries profound implications. The successful knowledge transfer from microscopic pathology slides to macroscopic surface lesions strongly indicates that the

TABLE V
IN-DOMAIN ACCURACY OF PATHVLM-R1 AND ITS PREDECESSOR MODELS ACROSS STAGED TRAINING PIPELINE (ACC. REPRESENTS ACCURACY).

Model	Accuracy (%)
Qwen2.5-VL-3B	44.50
Qwen2.5-VL-7B	51.60
Qwen2.5-VL-32B	54.20
PathVLM-α-3B	54.60
PathVLM-β-3B	56.90
PathVLM-γ-3B	54.40
PathVLM-δ-3B	52.10
PathVLM-ϵ-3B	49.60
PathVLM-α-7B	57.60
PathVLM-β-7B	<u>63.90</u>
PathVLM-γ-7B	59.90
PathVLM-δ-7B	52.90
PathVLM-ϵ-7B	56.10
Ours (PathVLM-R1)	**65.50**

model has learned fundamental pathological principles rather than superficial image features. This validates our staged training paradigm as a parameter-efficient and effective pathway for developing robust, interpretable foundational models poised for diverse clinical applications.

Analysis of Out-of-domain Data Results Despite these promising results, we acknowledge several limitations in our current work. Our process reward mechanism is contingent upon the evaluative capabilities of GPT-4o. Although we confirmed its scoring consistency through independent runs, this external model is not a certified pathology expert and may harbor inherent biases or knowledge gaps, which could introduce subtle noise into our reward signal. Secondly, our study was primarily conducted using the PathMMU dataset, which is centered on multiple-choice question-answering tasks. While this is a valuable benchmark, the model's proficiency in more open-ended, generative tasks—such as composing a full diagnostic report from a slide—or its ability to handle rare diseases not represented in the training data remains unverified. Lastly, while PathVLM-R1 demonstrated excellent generalization to the closely related domain of dermoscopy, its performance gains were more modest on modalities with disparate imaging principles, such as Chest CT and Retinal OCT, indicating that the boundaries of its cross-modal knowledge transfer still need to be expanded.

Future Research Directions Building upon our findings and acknowledging these limitations, we identify several key avenues for future research. To mitigate the reliance on a single AI evaluator, we plan to refine the reward model by incorporating feedback from human pathology experts, thereby creating a more accurate and professionally aligned human-in-the-loop system. We also intend to expand the scope of our work by training on more diverse datasets, particularly those including Whole Slide Images (WSIs), and by tackling more complex tasks such as automated report generation and patient prognosis prediction. To improve generalization, we will explore methods for strategically integrating a wider variety of medical imaging modalities into the training process. Ultimately, our goal is to advance this research toward prospective clinical validation, deploying PathVLM-R1 in real-world workflows to rigorously assess its utility and safety as a collaborative tool for pathologists.

VI. Conclusion

This paper introduces PathVLM-R1, a visual-language reasoning model tailored for pathology images, capable of effectively enhancing reasoning capability and diagnostic accuracy. By employing a combined strategy of supervised fine-tuning and reinforcement learning based on Qwen2.5-VL-7B-Instruct, we achieved the infusion of pathological knowledge and structured construction of the reasoning process. Experimental results demonstrate that PathVLM-R1 significantly outperforms traditional models across various medical imaging domains, particularly excelling in pathology image question-answering tasks. Its dual-reward mechanism effectively integrates interpretability of the process with result accuracy, showcasing strong adaptability and generalization capabilities.

The study not only provides a new perspective for automated reasoning in pathology image diagnostics but also lays the groundwork for multi-scenario applications across other medical imaging modalities. Looking ahead, PathVLM-R1 is poised to further enhance performance in diverse application domains through richer data training and optimization strategies, advancing the automation of medical image analysis. We believe that as algorithms, computational power, and data evolve, PathVLM-R1 can be extended to broader clinical scenarios, contributing to the development of precision medicine and intelligent healthcare.

References

[1] Z. Huang, F. Bianchi, M. Yuksekgonul, T. J. Montine, and J. Zou, "A visual–language foundation model for pathology image analysis using medical Twitter," *Nature Medicine*, vol. 29, no. 9, pp. 2307–2316, 2023.
[2] M. Y. L. et al., "A multimodal generative AI copilot for human pathology," *Nature*, vol. 634, no. 8033, pp. 466–473, 2024.
[3] J. Li, R. Selvaraju, A. Gotmare, S. Joty, C. Xiong, and S. C. H. Hoi, "Align before fuse: Vision and language representation learning with momentum distillation," in *Proceedings of the Advances in Neural Information Processing Systems*, 221, pp. 9694–9705.
[4] C. Li, C. Wong, S. Zhang, N. Usuyama, H. Liu, J. Yang, T. Naumann, H. Poon, and J. Gao, "LLaVA-Med: Training a large language-and-vision assistant for biomedicine in one day," in *Proceedings of the Advances in Neural Information Processing Systems*, 2023, pp. 28 541–28 564.
[5] Y. Xin, G. C. Ates, K. Gong, and W. Shao, "Med3DVLM: An efficient vision-language model for 3D medical image analysis," arXiv:2403.07826, 2024.
[6] D. G. et al., "DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning," arXiv:2501.12948, 2025.
[7] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," arXiv:1707.06347, 2017.
[8] H. Lightman, V. Kosaraju, Y. Burda, H. Edwards, B. Baker, T. Lee, J. Leike, J. Schulman, I. Sutskever, and K. Cobbe, "Let's verify step by step," in *Proceedings of the International Conference on Learning Representations*, 2023.
[9] H. L. et al., "RLAIF vs. RLHF: Scaling reinforcement learning from human feedback with AI feedback," arXiv:2309.00267, 2023.
[10] Z. S. et al., "DeepSeekMath: Pushing the limits of mathematical reasoning in open language models," arXiv:2402.03300, 2024.
[11] A. G. et al., "The Llama 3 herd of models," arXiv:2407.21783, 2024.
[12] G. T. et al., "Gemma 2: Improving open language models at a practical size," arXiv:2408.00118, 2024.
[13] S. B. et al., "Qwen2.5-VL technical report," arXiv:2502.13923, 2025.
[14] J. Bai, S. Bai, S. Yang, S. Wang, S. Tan, P. Wang, J. Lin, C. Zhou, and J. Zhou, "Qwen-VL: A versatile vision-language model for understanding, localization, text reading, and beyond," arXiv:2308.12966, 2023.
[15] H. L. et al., "DeepSeek-VL: Towards real-world vision-language understanding," arXiv:2403.05525, 2024.
[16] H. Liu, C. Li, Q. Wu, and Y. J. Lee, "Visual instruction tuning," in *Proceedings of the Advances in Neural Information Processing Systems*, 2023, pp. 34 892–34 916.
[17] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," in *Proceedings of the Advances in Neural Information Processing Systems*, 2017, pp. 5998–6008.
[18] H. Yang, H. Lu, X. Zeng, Y. Liu, X. Zhang, H. Yang, Y. Zhang, S. Huang, Y. Wei, and W. Lam, "Stephanie: Step-by-step dialogues for mimicking human interactions in social conversations," *arXiv preprint arXiv:2407.04093*, 2024.
[19] P. R. et al., "CheXNet: Radiologist-level pneumonia detection on chest x-rays with deep learning," arXiv:1711.05225, 2017.
[20] Y. Su, D. Li, and X. Chen, "Lung nodule detection based on faster R-CNN framework," *Computer Methods and Programs in Biomedicine*, vol. 200, p. 105866, 2021.
[21] S. Eslami, C. Meinel, and G. D. Melo, "PubMedCLIP: How much does CLIP benefit visual question answering in the medical domain?" in *Proceedings of the Findings of the Association for Computational Linguistics: EACL*, 2023, pp. 1181–1193.

[22] K. Yan, Y. Liu, J. Li, J. Cao, and D. Zhou, "Pathological image segmentation of breast cancer via template matching," *IEEE Journal of Biomedical and Health Informatics*, pp. 1–12, 2025.
[23] Z. Hu, J. Wang, Q. Gao, Z. Wu, H. Xu, Z. Guo, J. Quan, L. Zhong, M. Du, T. Tong, and G. Chen, "Weakly supervised classification for nasopharyngeal carcinoma with Transformer in whole slide images," *IEEE Journal of Biomedical and Health Informatics*, vol. 28, no. 12, pp. 7251–7262, 2024.
[24] J. Zhang, J. Huang, S. Jin, and S. Lu, "Vision-language models for vision tasks: A survey," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2024.
[25] J. P. et al., "MedVLM-R1: Incentivizing medical reasoning capability of vision-language models (VLMs) via reinforcement learning," arXiv:2502.19634, 2025.
[26] X. Feng, D. Wang, Q. Pan, M. Yan, X. Liu, Y. Shen, L. Fang, G. Cai, and G. Ning, "Reinforcement learning model for managing noninvasive ventilation switching policy," *IEEE Journal of Biomedical and Health Informatics*, vol. 27, no. 8, pp. 4120–4130, 2023.
[27] P. F. Christiano, J. Leike, T. Brown, M. Martic, S. Legg, and D. Amodei, "Deep reinforcement learning from human preferences," in *Proceedings of the Advances in Neural Information Processing Systems*, 2017.
[28] J. A. et al., "GPT-4 technical report," arXiv:2303.08774, 2023.
[29] G. Sheng, C. Zhang, Z. Ye, X. Wu, W. Zhang, R. Zhang, Y. Peng, H. Lin, and C. Wu, "HybridFlow: A flexible and efficient RLHF framework," arXiv:2409.19256, 2024.
[30] H. Lee, S. Phatale, H. Mansoor, K. R. Lu, T. Mesnard, J. Ferret, C. Bishop, E. Hall, V. Carbune, and A. Rastogi, "RLAIF: Scaling reinforcement learning from human feedback with AI feedback," arXiv:2309.00267, 2023.
[31] L. O. et al., "Training language models to follow instructions with human feedback," in *Proceedings of the Advances in Neural Information Processing Systems*, 2022, pp. 27 730–27 744.
[32] Y. B. et al., "Constitutional AI: Harmlessness from AI feedback," arXiv:2212.08073, 2022.
[33] X. Pang, S. Tang, R. Ye, Y. Xiong, B. Zhang, Y. Wang, and S. Chen, "Self-alignment of large language models via monopolylogue-based social scene simulation," arXiv:2402.05699, 2024.
[34] Y. S. et al., "PathMMU: A massive multimodal expert-level benchmark for understanding and reasoning in pathology," in *Proceedings of the European Conference on Computer Vision*, 2024, pp. 56–73.
[35] Y. Sunne, "Chest CT-Scan images Dataset," [Online]. Available: https://tianchi.aliyun.com/dataset/93929, 2021.
[36] V. R. et al., "A patient-centric dataset of images and metadata for identifying melanomas using clinical context," *Scientific Data*, vol. 8, no. 1, p. 34, 2021.
[37] M. Subramanian, K. Shanmugavadivel, O. S. Naren, K. Premkumar, and K. Rankish, "Classification of retinal OCT images using deep learning," in *Proceedings of the International Conference on Computing, Communication and Informatics*, 2022, pp. 1–7.
[38] A. T. Platform, "Diabetic Retinopathy Arranged Dataset," [Online]. Available: https://tianchi.aliyun.com/dataset/93926, 2023, accessed: 2021-03-10.
[39] Y. Hu, T. Li, Q. Lu, W. Shao, J. He, Y. Qiao, and P. Luo, "OmniMedVQA: A new large-scale comprehensive evaluation benchmark for medical LVLM," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 22 170–22 183.